SE 226 – LAB#2 2025-2026 SPRING

Aim: Flow control: selection statements , looping statements.

General Instructions

Create a Python project containing **lab2.py** and a C++ source file **lab2.cpp**. Complete all tasks in **both** programming languages. The problem names are provided as references — feel free to look them up online if you need further clarification.

Task 1 (Digital Root Sequence)

Take a **positive integer greater than 9** from the user.

Repeatedly replace the number with the **sum of its digits** until the value becomes a **single-digit number**.

Print each intermediate value in the sequence on the same line separated by " → ".

Finally, print the **final value** and the **total number of steps** taken.

*Hint: Use a **while loop** to repeat the process until the number becomes a single digit.*
*To compute the **sum of digits**, repeatedly extract digits using division and modulo operations.*

SAMPLE OUTPUT(n=987)

```
Please enter a positive integer greater than 9: 987

987 → 24 → 6
Final value: 6
Total steps: 2
```

SAMPLE OUTPUT(n=199999)

```
Please enter a positive integer greater than 9: 199999

199999 → 46 → 10 → 1
Final value: 1
Total steps: 3
```

Task 2 (FizzBuzz Counter)

Take a positive integer **n** from the user (between **10 and 100** inclusive). If the user enters a value outside this range, keep asking until a valid value is given. Then iterate through all integers from **1 to n** and apply the following rules:

• If the number is divisible by both 3 and 5 → print **FizzBuzz**
• Else if divisible by 3 → print **Fizz**

• Else if divisible by 5 → print **Buzz**

• If the number is divisible by 7 → **skip it entirely** (use **continue**)

• Otherwise → print the number itself

After the loop, print a summary: how many **Fizz**, **Buzz**, and **FizzBuzz** values were printed.

Hint: Use a for loop with range(), elif chain, and the continue statement for the divisible-by-7 skip. Use a separate while loop for input validation.

SAMPLE OUTPUT (n=20)

```
Please enter a number between 10 and 100: 5
Invalid input. Please enter a number between 10 and 100: 110
Invalid input. Please enter a number between 10 and 100: 20
1
2
Fizz
4
Buzz
Fizz
(7 is skipped)
8
Fizz
Buzz
11
Fizz
13
(14 is skipped)
FizzBuzz
16
17
Fizz
19
Buzz
--- Summary ---
Fizz count : 5
Buzz count : 3
FizzBuzz count: 1
```

Task 3

Take a number from the user between 3 and 9 (inclusive). Print the following pattern using at most two loop structures.

*Note: The pattern first increases from **1 up to n**, then decreases from **n−1 back to 1**. For each row k, print numbers from **1 to k.***

SAMPLE OUTPUT (n=4)

```
Please enter a number between 3 and 9: 4
1
12
123
1234
123
12
1
```

SAMPLE OUTPUT (n=7)

```
Please enter a number between 3 and 9: 7
1
12
123
1234
12345
123456
1234567
123456
12345
1234
123
12
1
```

*Hint: For each row i (1 ≤ i ≤ 2n−1), determine the current **level** k = n − |n − i|. Print the numbers from **1 to k** on the same line..*

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*Please complete all tasks in both **Python and C++**. Upload your `lab2.py` and `lab2.cpp` files into your **GitHub account**, then share the link to your repository/folder via **Blackboard**.*